

# Vrajesh Sharma

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## Summary

IEEE-published researcher and pre-final year B.Tech (AI & ML) student, currently interning at IIT Patna on medical signal digitization. Skilled in end-to-end deep learning pipelines, LLM-based multi-agent systems, and RAG architectures, with deployed projects. Machine Learning Head at ASPDC, with experience teaching 150+ students across ML and DL programs.

## Education

**Adani University - Faculty of Engineering Science and Technology**  
B.Tech in Computer Science Engineering (AI & ML) — CGPA: 8.24

July 2023 – Expected May 2027  
Ahmedabad, Gujarat

## Experience

**Research Intern, Indian Institute of Technology - Patna**

Hybrid   May 2025 – Present

- Developing medical graph digitization pipelines at IIT Patna to extract structured numerical data from clinical signal images, contributing to research in healthcare AI.

**Machine Learning Head, ASPDC**

Ahmedabad   Aug 2025 – Present

- Leading a team of 5 Machine Learning Leads to design and deliver ML initiatives and learning programs for the club.
- Organized and taught the **ML in One Month** and **DL in One Month** programs for **150+** students, covering end-to-end ML pipelines, deep learning (ANN, CNN, RNN, LSTM), and model deployment workflows.

**Machine Learning Lead, ASPDC**

Ahmedabad   Aug 2024 – Aug 2025

- Conducted an ML-101 workshop introducing Linear Regression and fundamentals of supervised learning.

## Projects

**TumorTrace** — *PyTorch, U-Net, FastAPI, React, BraTS2020*

[Source Code](#)   Mar 2026

- Built an end-to-end brain tumor segmentation system using a 2D U-Net trained on 57K+ MRI slices (45K train / 11K validation), achieving  $>0.87$  Dice score on validation.
- Developed a full-stack ML application with FastAPI REST APIs and React, generating real-time tumor overlays, class-wise segmentation, and automated diagnostic reports with tumor area statistics.

**LekhAI** — *HTML, Flask, Gemini, DuckDuckGo, Render*

[Live Deployment](#)   [Source Code](#)   Nov 2025

- Built a multi-agent AI system for automated content generation using LLM orchestration and real-time web retrieval, reducing manual research and drafting effort by automating end-to-end blog generation.
- Designed a scalable agent architecture (Researcher, Strategist, Writer, Editor) supporting streaming responses, Markdown rendering, and structured multi-step reasoning workflows.

**Sarathi** — *Gemini, React, TypeScript, Flask*

[Live Deployment](#)   [Source Code](#)   Dec 2025

- Developed and deployed a RAG-based AI system leveraging vector search over curated datasets to deliver context-aware and citation-backed responses.
- Designed modular pipelines for retrieval, reasoning, and response generation, improving response relevance and grounding through knowledge-based augmentation.

**Swabhav** — *TensorFlow, Flask, Joblib, Scikit-learn, HTML*

[Live Deployment](#)   [Source Code](#)   Nov 2025

- Engineered and deployed a deep neural network-based personality classification system trained on 43K+ samples, achieving 89.06% test accuracy with strong generalization.
- Evaluated multiple models and selected optimal architecture based on accuracy, F1-score, and overfitting gap analysis.

## Achievements

- First Author** — Published "*Empowering Text Classification with Agentic AI: A Systematic Review*" in IEEE Xplore, presented at the 2nd International IEEE Conference on Artificial Intelligence and Machine Vision.   [View Paper](#)
- IEEE Member** and Volunteer at **IEEE AISYWLC 2025**.

## Skills

**Machine Learning & AI:** Regression, Classification, Feature Engineering, Hyperparameter Tuning, Model Deployment

**Deep Learning & GenAI:** Neural networks, LLMs, RAG, Vector Databases, Multi-Agent Systems, Fine-tuning

**Libraries & Frameworks:** TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, Matplotlib, LangChain, CrewAI, Flask, FastAPI

**Languages:** Python, SQL, C, C++

**Databases:** MySQL, PostgreSQL, Pinecone, ChromaDB, Supabase

**Tools & Platforms:** Git, GitHub, Docker, REST APIs, VS Code, Vercel, Render, n8n